Lecture 4

Copies and moves

Objectives. *Lecture 3 found the state of part of a register. This lecture asks:* what can be done with an unknown quantum state? *It cannot be copied (Theorem 4.1). It can be moved: a Bell pair, a Bell-basis measurement and two bits teleport a qubit, leaving nothing behind that depends on it (Theorem 4.3, the main result); nothing about it reaches the receiver before either bit (Theorem 4.4). And a reading of wires used afterwards only as controls can wait until the end (Theorem 4.5).*

By the end of this lecture you will be able to:

- Prove that no gate clones two states unless they are orthogonal or equal up to a phase, and show that a CNOT copies $|0\rangle$ and $|1\rangle$ but not $|+\rangle$.
- Compute the teleportation circuit: each outcome has probability $\frac{1}{4}$, and $Z^{m_1}X^{m_2}$, with X first, returns the input exactly.
- Prove that before either bit arrives the receiver's qubit is in $\frac{1}{2}\mathbb{1}$, whatever the input.
- Prove that a reading followed by gates it controls gives the same joint statistics as the controlled gates followed by the reading, and that the hypothesis is needed.

4.1 No-cloning

Lecture 1 copied bits freely: $\mathrm{CNOT}(a, 0) = (a, a)$ in (1.7) is FAN, and Bennett's construction copies its answer before it uncomputes (Example 1.10). Is there a gate that takes an unknown state ψ and a blank register to two copies of ψ? It may use a workspace, and leave in it whatever it likes.

Theorem 4.1 (No-cloning). *Let A and B be registers of n qubits each and E a further register; let ω be a state of BE and U a gate on ABE. If two states ψ and φ of A are* cloned *by U, that is,*

$$U(\psi \otimes \omega) = \psi \otimes \psi \otimes e_\psi,$$
$$U(\varphi \otimes \omega) = \varphi \otimes \varphi \otimes e_\varphi$$

for some states e_ψ and e_φ of E, then $|\langle\psi|\varphi\rangle|$ is 0 or 1: the two states are orthogonal, or $\varphi = e^{i\gamma}\psi$ for a real γ.

Proof. A gate keeps inner products: $\langle Uv|Uw\rangle = \langle v|U^\dagger U|w\rangle = \langle v|w\rangle$. By (3.1), the two inputs have the inner product $\langle\psi|\varphi\rangle\langle\omega|\omega\rangle = \langle\psi|\varphi\rangle$, and the two outputs $\langle\psi|\varphi\rangle^2 \langle e_\psi|e_\varphi\rangle$. So $c = |\langle\psi|\varphi\rangle|$ has $c = c^2|\langle e_\psi|e_\varphi\rangle| \le c^2$, by the Cauchy–Schwarz inequality, and $0 \le c \le 1$ by the same inequality; hence c is 0 or 1. If $c = 1$, put $\lambda = \langle\psi|\varphi\rangle$: then $\|\varphi - \lambda\psi\|^2 = 1 - |\lambda|^2 = 0$, so $\varphi = \lambda\psi$ with $|\lambda| = 1$. ∎

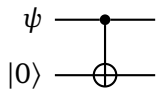
The workspace E may hold whatever a circuit of gates uses, and ω may be entangled. A gate on AB alone that clones up to a phase, $U(\psi \otimes s) = e^{i\alpha}\psi \otimes \psi$, is the case of E idle in a

state e, with $e_\psi = e^{i\alpha}e$. Cloning is a property of rays: if U clones ψ, it clones $e^{i\gamma}\psi$, with $e^{-i\gamma}e_\psi$. That is why the conclusion is about $|\langle\psi|\varphi\rangle|$. Nielsen and Chuang (Box 12.1, p. 532) ask for $\psi\otimes\psi$ exactly and conclude $\langle\psi|\varphi\rangle \in \{0,1\}$; read so, no gate clones both ψ and $-\psi$, which are one state (Theorem 3.7).

In particular, no gate clones both $|0\rangle$ and $|+\rangle$, since $\langle 0|+\rangle = 1/\sqrt{2}$, so none clones every state of a qubit. That weaker statement is what Griffiths and Schroeter ((12.41)–(12.45)) and Watrous (p. 87) prove: by linearity, a machine that clones two states sends their superpositions to the wrong vectors. Griffiths and Schroeter motivate it by relativity, since a cloner would let a measurement on one particle of an entangled pair send a message faster than light (p. 459). The proof above uses no relativity, only that gates keep inner products. The alternative "orthogonal" does occur, as the next section shows.

4.2 The CNOT guard

Example 4.2 (QQ: CNOT is no copier). On a blank target the CNOT is FAN: $\mathrm{CNOT}\,|j\rangle\,|0\rangle = |j\rangle\,|j\rangle$. On $\psi = a\,|0\rangle + b\,|1\rangle$, by linearity,



$$= a\,|00\rangle + b\,|11\rangle\,, \tag{4.1}$$

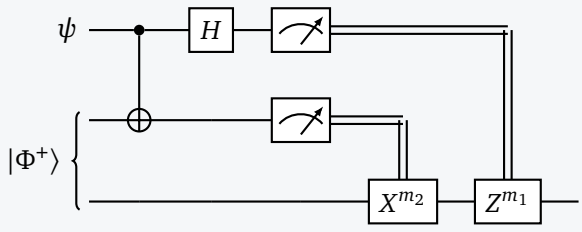
an equality of vectors, while $\psi\otimes\psi = a^2\,|00\rangle + ab\,|01\rangle + ab\,|10\rangle + b^2\,|11\rangle$. The amplitude of $|01\rangle$ is 0 on the left, so the CNOT clones ψ, even up to a phase, only if $ab = 0$: on the rays of $|0\rangle$ and $|1\rangle$. On $|+\rangle$ it makes Φ^+ (Example 3.11), which is entangled, and whose halves are in $\frac{1}{2}\mathbb{1}$, not in $|+\rangle$.

So the CNOT copies a basis, as FAN does; on Lecture 1's fair coin FAN gives 00 or 11, not two independent coins. Lecture 10 encodes ψ as $a\,|000\rangle + b\,|111\rangle$ by two such CNOTs (Nielsen and Chuang, p. 427): an encoding, not three copies.

4.3 Teleportation

A state cannot be copied; it can be moved. Alice holds a qubit in an unknown state ψ, and shares with Bob the Bell state Φ^+, made beforehand by the circuit (3.6), one qubit each. She can send Bob bits.

Theorem 4.3 (Teleportation). *Let wire 1 carry a qubit in a state ψ, and wires 2–3 the Bell state Φ^+; wires 1 and 2 are Alice's, wire 3 is Bob's. Alice applies a CNOT from wire 1 to wire 2 and then H to wire 1, and reads wires 1 and 2, with the outcomes m_1 and m_2; Bob applies X^{m_2} and then Z^{m_1} to wire 3:*



Then, for every ψ:

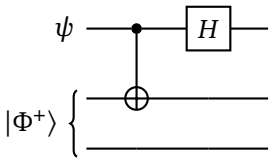
(i) each pair $m_1 m_2$ has probability $\frac{1}{4}$;

(ii) the reading leaves wires 1–2 in $|m_1 m_2\rangle$, and Bob's gates, the operator $Z^{m_1}X^{m_2}$, leave wire 3 in ψ exactly.

Proof. Write $\psi = a\,|0\rangle + b\,|1\rangle$. Then $\psi \otimes \Phi^+ = (a\,|000\rangle + a\,|011\rangle + b\,|100\rangle + b\,|111\rangle)/\sqrt{2}$; the CNOT turns the last two terms into $b\,|110\rangle + b\,|101\rangle$, and H on wire 1 sends $|0x\rangle$ to $(|0x\rangle + |1x\rangle)/\sqrt{2}$ and $|1x\rangle$ to $(|0x\rangle - |1x\rangle)/\sqrt{2}$. Collected by the strings m_1m_2 of wires 1–2, the state is

$$\begin{aligned} &\tfrac{1}{2}\,|00\rangle\,(a\,|0\rangle + b\,|1\rangle) \\ +\,&\tfrac{1}{2}\,|01\rangle\,(a\,|1\rangle + b\,|0\rangle) \\ +\,&\tfrac{1}{2}\,|10\rangle\,(a\,|0\rangle - b\,|1\rangle) \\ +\,&\tfrac{1}{2}\,|11\rangle\,(a\,|1\rangle - b\,|0\rangle). \end{aligned}$$

Since $a\,|1\rangle + b\,|0\rangle = X\psi$, $a\,|0\rangle - b\,|1\rangle = Z\psi$ and $a\,|1\rangle - b\,|0\rangle = XZ\psi$, this says



$$= \frac{1}{2}\sum_{m_1,m_2} |m_1m_2\rangle \otimes X^{m_2}Z^{m_1}\psi, \tag{4.2}$$

an equality of vectors. Reading wires 1–2 reads part of the register: by (3.2) it gives m_1m_2 with probability $\left\|\frac{1}{2}X^{m_2}Z^{m_1}\psi\right\|^2 = \frac{1}{4}$, since X and Z are unitary, and leaves $|m_1m_2\rangle \otimes X^{m_2}Z^{m_1}\psi$. Bob's gates make this $|m_1m_2\rangle \otimes Z^{m_1}X^{m_2}X^{m_2}Z^{m_1}\psi = |m_1m_2\rangle \otimes \psi$, since $X^2 = Z^2 = \mathbb{1}$. ∎

Alice's part is a measurement in the Bell basis. With U the circuit of (3.6), H and then a CNOT, $U^\dagger$ is a CNOT and then H; so by Proposition 3.6 her CNOT, H and reading give m_1m_2 with the probability of the measurement of wires 1–2 in the basis $\{U\,|m_1m_2\rangle\}$ of the four Bell states, of which $U\,|00\rangle = \Phi^+$. Without the final U, they leave $|m_1m_2\rangle$: Alice's qubits end in a computational-basis state, not in a Bell state (the circuit of Nielsen and Chuang's Ex. 4.33).

What Alice is left with, the bits m_1m_2 and the state $|m_1m_2\rangle$, has the same distribution for every ψ: nothing that depends on the input stays with her. Nothing travels to Bob but the two bits; his own qubit, wire 3, ends in ψ. The order of his gates matters: $X^{m_2}Z^{m_1}$, Z first, would leave $X^{m_2}Z^{m_1}X^{m_2}Z^{m_1}\psi = (-1)^{m_1m_2}\psi$, the right ray (Theorem 3.7) but $-\psi$ on the outcome 11. Nielsen and Chuang (pp. 26–28) use our labels: their outcome 01, which calls for X, is $m_1 = 0$, $m_2 = 1$. Watrous (pp. 96–101) writes kets in the order Bob, Alice's half, input, and his outcome pair ab is our m_2m_1: his 01 calls for Z.

4.4 No signalling

Bob's gates need m_1m_2, and the two bits reach him no faster than light. Can he learn anything about ψ from his own qubit before either arrives?

> **Theorem 4.4** (No signalling in teleportation)**.** *In the circuit of Theorem 4.3, until Bob learns either bit, his qubit is in the reduced state $\frac{1}{2}\mathbb{1}$ at every step, for every input ψ: at the start, after Alice's CNOT, after her H, and after her reading.*

Proof. After the H the register is in the state χ of (4.2), which is $\sum_m |m_1m_2\rangle \otimes \chi_m$ with $\chi_m = \frac{1}{2}X^{m_2}Z^{m_1}\psi$. By Corollary 3.10, with the parts exchanged (Alice's wires traced out in the basis $|m_1m_2\rangle$), Bob's state is

$$\begin{aligned} \rho_B &= \textstyle\sum_m |\chi_m\rangle\langle\chi_m| \\ &= \tfrac{1}{4}\textstyle\sum_m X^{m_2}Z^{m_1}\rho Z^{m_1}X^{m_2}, \end{aligned}$$

with $\rho = |\psi\rangle\langle\psi|$. For a 2×2 matrix ρ, $\frac{1}{2}(\rho + Z\rho Z)$ is its diagonal D, and $\frac{1}{2}(D + XDX)$ is $\frac{1}{2}(\operatorname{Tr} D)\mathbb{1}$; so the sum is $\frac{1}{2}(\operatorname{Tr}\rho)\mathbb{1} = \frac{1}{2}\mathbb{1}$. After the reading, Bob's qubit is in $X^{m_2}Z^{m_1}\psi$ with probability $\frac{1}{4}$ for each $m_1 m_2$; knowing neither bit, he holds the mixture of these states, which is the same matrix. Before the H, and before the CNOT, the register is in $W\chi$ for a gate $W = V\otimes\mathbb{1}$ with V on wires 1–2. For every 2×2 matrix M, the interchange law gives $W^\dagger(\mathbb{1}\otimes M)W = V^\dagger V\otimes M = \mathbb{1}\otimes M$, so $\operatorname{Tr}\big(W\sigma W^\dagger(\mathbb{1}\otimes M)\big) = \operatorname{Tr}\big(\sigma(\mathbb{1}\otimes M)\big)$ for $\sigma = |\chi\rangle\langle\chi|$, and by the uniqueness in Theorem 3.9 Bob's state is unchanged. ∎

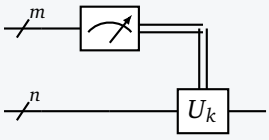
So before either bit arrives, every measurement of Bob's qubit has the statistics of $\frac{1}{2}\mathbb{1}$, the same for every ψ: in every basis, each outcome has probability $\frac{1}{2}$ (§ 3.5). Nothing about ψ reaches Bob before a bit does. The bits alone carry nothing about ψ either: by Theorem 4.3 (i) they are uniform for every ψ. Only together with Bob's qubit do they give ψ. Nielsen and Chuang (p. 108) compute $\frac{1}{2}\mathbb{1}$ after Alice's reading; it holds at every step.

4.5 Deferred measurement

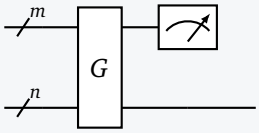
Bob's gates are *classically controlled*: which one acts depends on the outcome of a reading. Can the reading wait until the end?

Theorem 4.5 (Deferred measurement)**.** *Let AB be a register, with A of m qubits, and for each $k\in\{0,1\}^m$ let U_k be a gate on B; put $G = \sum_k |k\rangle\langle k|\otimes U_k$. On every state χ of AB the procedures*

(a) read A, then apply U_k to B, where k is the outcome:



(b) apply G, then read A:



give each outcome k with the same probability p_k and, if $p_k > 0$, leave the same state. So all later steps give the same joint statistics.

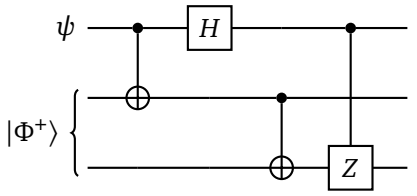
Proof. G is a gate, since $G^\dagger G = \sum_k |k\rangle\langle k|\otimes U_k^\dagger U_k = \mathbb{1}$. Let $\Pi_k = |k\rangle\langle k|\otimes\mathbb{1}$ be the projectors of the reading of A; by the interchange law, $G\Pi_k = |k\rangle\langle k|\otimes U_k = \Pi_k G$. In (a), the reading gives k with $p_k = \|\Pi_k\chi\|^2$ and, if $p_k > 0$, leaves $\Pi_k\chi/\sqrt{p_k} = |k\rangle\otimes\varphi_k$ by (3.2); then U_k makes it $|k\rangle\otimes U_k\varphi_k = G(|k\rangle\otimes\varphi_k)$. In (b), the reading gives k with $\|\Pi_k G\chi\|^2 = \|G\Pi_k\chi\|^2 = p_k$, since G is unitary, and, if $p_k > 0$, leaves $\Pi_k G\chi/\sqrt{p_k} = G\Pi_k\chi/\sqrt{p_k} = G(|k\rangle\otimes\varphi_k)$, as (a) does. ∎

After the reading, A holds $|k\rangle$, on which a gate $\sum_j |j\rangle\langle j|\otimes V_j$, one that touches A only as a control, acts as V_k: as the gate classically controlled by the outcome. Gates on B alone have this form, as do controlled gates with the control in A and the target in B, and products of these. So if every step after a reading is such a gate, the reading can be moved to the end, and classically controlled gates become controlled ones.

The hypothesis bites: H does not commute with the Π_k. Read $|+\rangle$, apply H and read again: 0 and 1, with probability $\frac{1}{2}$ each. Without the first reading, $H|+\rangle = |0\rangle$ reads 0 with

certainty (Theorem 3.7). The runs share joint statistics, not states: before its reading, (b) holds a superposition of the branches of (a). Nielsen and Chuang's principle (p. 186) omits the hypothesis; their Ex. 4.35 (p. 188) has it.

Example 4.6 (QQ plus an input: teleportation, deferred)**.** Theorem 4.3 is the case A = wires 1–2, B = wire 3, $U_{m_1m_2} = Z^{m_1}X^{m_2}$: G is a CNOT from wire 2 to wire 3, then a controlled Z from wire 1 to wire 3, which acts as Z^{m_1} on wire 3. By (4.2) it takes χ to $\frac{1}{2}\sum_m |m_1m_2\rangle \otimes \psi$:



$$= |+\rangle\,|+\rangle \otimes \psi, \tag{4.3}$$

an equality of vectors. Read at the end, wires 1–2 give each m_1m_2 with probability $\frac{1}{4}$, and wire 3 holds ψ. It is no protocol for Alice and Bob apart: its last two gates join her wires to his; Lecture 6 uses this form for the coins of a probabilistic algorithm.

Summary.

- *Copies:* no gate clones two states unless they are orthogonal or one ray (Theorem 4.1); a CNOT copies $|0\rangle$ and $|1\rangle$, not $|+\rangle$ (Example 4.2).
- *The main result:* a Bell pair, a Bell-basis reading and two bits teleport a qubit; each outcome has probability $\frac{1}{4}$, and $Z^{m_1}X^{m_2}$ returns the input exactly (Theorem 4.3).
- *No signalling:* until Bob learns either bit, his qubit is in $\frac{1}{2}\mathbb{1}$ (Theorem 4.4).
- *Deferred measurement:* a reading of wires used afterwards only as controls moves to the end, with the same joint statistics (Theorem 4.5); Lecture 6 uses it.
- *Next:* Lecture 5, questions to an oracle.

References

Nielsen and Chuang, pp. 24–28, 106–108, 185–188 (Ex. 4.33, 4.35), 427 and 532 (Box 12.1). Watrous (2025), pp. 87–89 and 96–101 (little-endian). Griffiths and Schroeter, pp. 459–460. W. K. Wootters and W. H. Zurek, Nature **299** (1982), 802–803; D. Dieks, Phys. Lett. A **92** (1982), 271–272. C. H. Bennett *et al.*, Phys. Rev. Lett. **70** (1993), 1895–1899.